

CM0246 Discrete Structures

Representing Graphs and Graph Isomorphism

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Semestre 2014-2

Preliminaries

Textbook

Kenneth H. Rosen ([1988] 2004). Matemática Discreta y sus Aplicaciones.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Graphs

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Simple Graphs

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Adjacent Vertices and Incident edges

Definition 1

Two vertices u and v in an undirected graph G are called **adjacent** in G if $\{u, v\}$ is an edge of G . If $e = \{u, v\}$, the edge e is called **incident** with the vertices u and v .

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Degrees of the Vertices

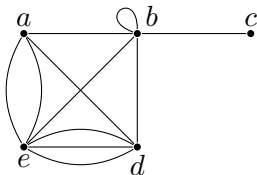
Definition

The **degree of a vertex** v in an undirected graph, denoted $\delta(v)$, is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex.

Degrees of the Vertices

Exercise

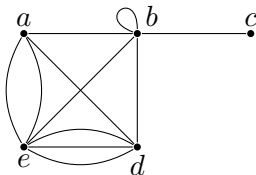
Find the degree of each vertex in the following graph:



Degrees of the Vertices

Exercise

Find the degree of each vertex in the following graph:



Solution

$$\delta(a) = 4, \delta(b) = \delta(e) = 6, \delta(c) = 1 \text{ and } \delta(d) = 5.$$

Vertex Degrees

Theorem 1 (the handshaking theorem, p. 511)

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Proof.

Each edge contributes **two** to the sum of the degrees of the vertices because an edge is **incident** with exactly two (possibly equal) vertices. This means that the sum of the degrees of the vertices is twice the number of edges. ■

Representing Graphs

- Adjacency matrices
- Incidence matrices

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$$a_{ij} = \begin{cases} 1, & \text{if } \{v_i, v_j\} \text{ is an edge of } G; \\ 0, & \text{otherwise.} \end{cases}$$

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Isomorphism of Graphs

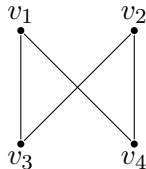
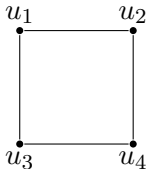
Definition

The simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are **isomorphic** if there exists a bijective function f from V_1 to V_2 with the property that u and v are adjacent in G_1 , if and only if, $f(u)$ and $f(v)$ are adjacent in G_2 , for all u and v in V_1 .

Isomorphism of Graphs

Example

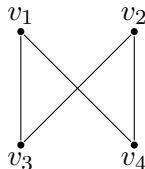
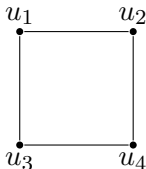
The following simple graphs are isomorphic.



Isomorphism of Graphs

Example

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The bijective function f preserves adjacency.

$$f : \{u_1, u_2, u_3, u_4\} \rightarrow \{v_1, v_2, v_3, v_4\}$$

$$f(u_1) = v_1, f(u_2) = v_4, f(u_3) = v_3 \text{ and } f(u_4) = v_2.$$

Isomorphism of Graphs

Remark

Determining whether two simple graphs are isomorphic is often difficult because if $|A| = |B| = n$ then

$$|\{ f : A \rightarrow B \mid f \text{ is a bijection} \}| = n!.$$

Isomorphism of Graphs

Definition

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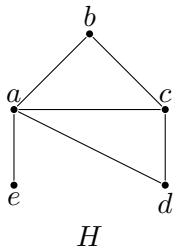
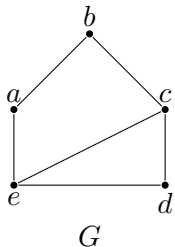
Remark

We can prove that two graphs are not isomorphic if we can find a graph invariant property that only one of the two graphs has.

Isomorphism of Graphs

Example

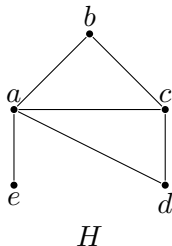
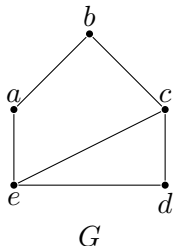
Are the following graphs isomorphic?



Isomorphism of Graphs

Example

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Solution

No. The graph H has a vertex of degree 1 but the graph G have no vertices of degree 1.

Isomorphism of Graphs

Definition

The **complementary graph** $\overline{G}$ of a simple graph G has the same vertices as G . Two (different) vertices are adjacent in $\overline{G}$ if and only if they are not adjacent in G .

Example

Whiteboard.

Isomorphism of Graphs

Definition

A simple graph G is called **self-complementary** if G and $\overline{G}$ are isomorphic.

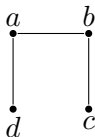
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Problem 50 (p. 529)

Is the given graph
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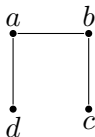
Isomorphism of Graphs

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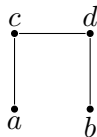
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Is the given graph self-complementary?



Yes! The complementary graph is given by the figure.



The isomorphism is $f(a) = c$, $f(b) = d$,
 $f(c) = b$ and $f(d) = a$.

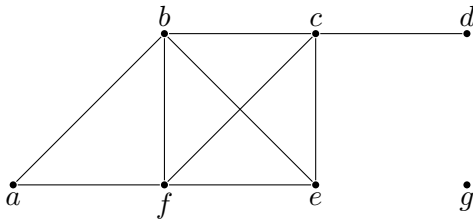
Isomorphism of Graphs

Definition

The **degree sequence** of a graph is the sequence of the degrees of the vertices of the graph in non-increasing order.

Example

For the graph in the figure, the degree sequence is 4, 4, 4, 3, 2, 1, 0.



Isomorphism of Graphs

Problem 69 (p. 530)

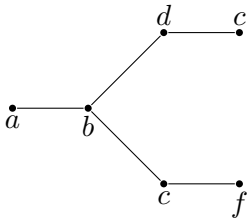
A counter-example for a purported isomorphism test is a pair of non-isomorphic graphs that the test fails to show that they are not isomorphic.

Find a counter-example for the test that checks the degree sequence in two graphs to make sure they agree.

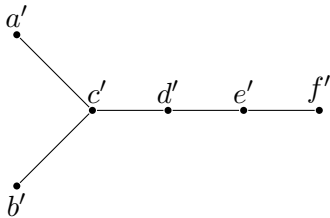
Isomorphism of Graphs

Solution

The degree sequence of both graphs is $3, 2, 2, 1, 1, 1$ but they are not isomorphic. In graph G , the vertex b has degree 3 and it is adjacent to two vertices of degree 2 and one vertex of degree 1. The graph H has no vertex with these properties.



G



H

Isomorphism of Graphs

Comparison of several time complexity functions

$f(n)$	10	50	100
$\log n$	2.3 sec	3.9 sec	4.6 sec
n	10 sec	50 sec	1.7 min
n^2	1.7 min	41.7 min	2.8 h
2^n	17.1 min	358.001 c	4×10^{20} c
3^n	16.4 h	2.3×10^{14} c	1.6×10^{38} c
$n!$	42 d	9.7×10^{54} c	3×10^{148} c

Isomorphism of Graphs

Algorithms for graph isomorphism

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vertices	10	100	1000	10000
$2^{\sqrt{n \log n}}$	27.8 sec	33.4 d	3.3×10^{15} c	7.3×10^{81} c

References

- David S. Johnson (2005). The NP-Completeness Column. ACM Transactions on Algorithms 1.1, pp. 160–176. DOI: [10.1145/1077464.1077476](https://doi.org/10.1145/1077464.1077476) (cit. on pp. 61, 62).
- Kenneth H. Rosen [1988] (2004). Matemática Discreta y sus Aplicaciones. Trans. by José Manuel Pérez Morales, Julio Moro Carreño, Ana Isabel Lías Quintero and Pedro Antonio Ramos Alonc. 5th ed. McGraw-Hill (cit. on p. 2).